

MATH 9 · ALBERTA CURRICULUM · NUMBER UNIT

Real World Problems

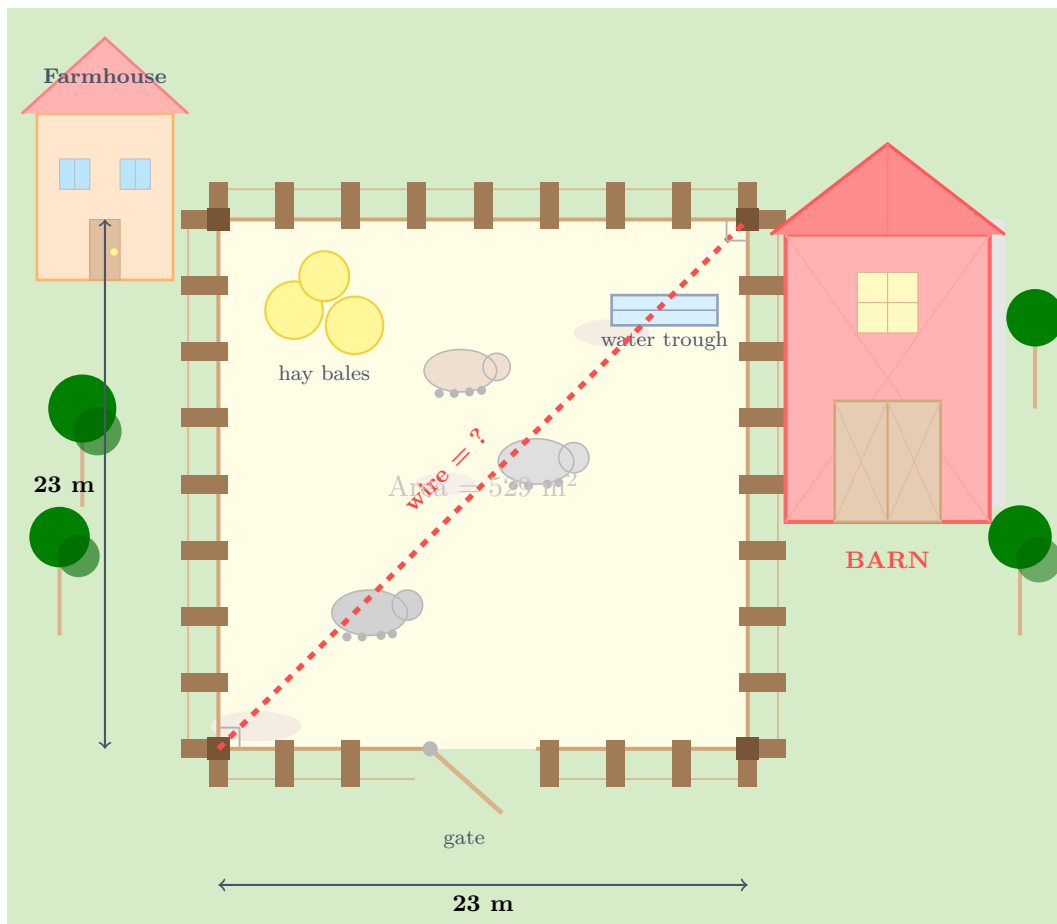
Square Roots · Rational Numbers · Powers & Exponents

Name: _____

Date: _____

Problem 1

Square Roots

**Problem 1: The Ranch Corral**

A rancher has built a square livestock corral with an area of 529 m^2 . She wants to run a diagonal support wire from one corner of the corral to the opposite corner.

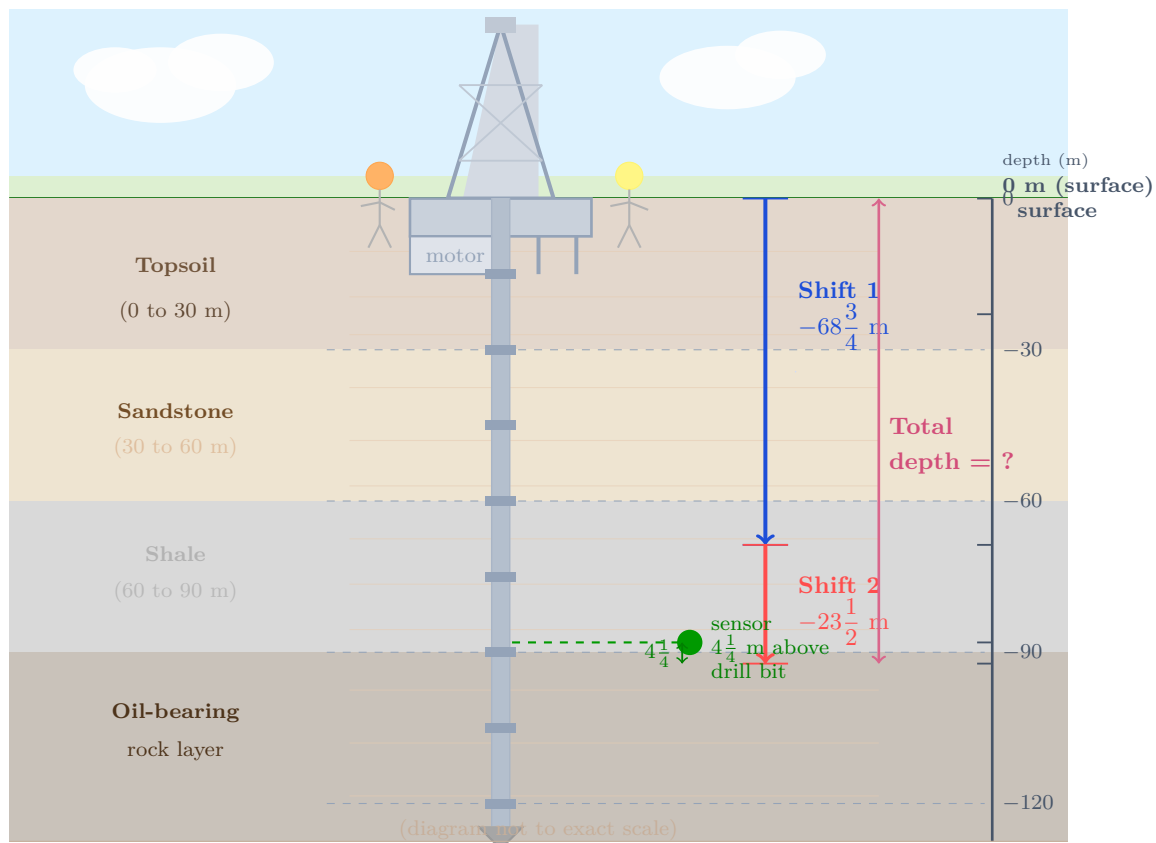
- What is the side length of the corral? Show your work.
- What is the minimum length of wire needed for the diagonal? Round to the nearest tenth of a metre.

(a)

(b)

Problem 2

Rational Numbers

**Problem 2: Going Deep**

An oil drilling crew starts at the surface (depth = 0 m). During the first shift, they drill down $68\frac{3}{4}$ m. During the second shift, they drill an additional $23\frac{1}{2}$ m deeper. A pressure sensor is attached to the drill pipe $4\frac{1}{4}$ m above the drill bit.

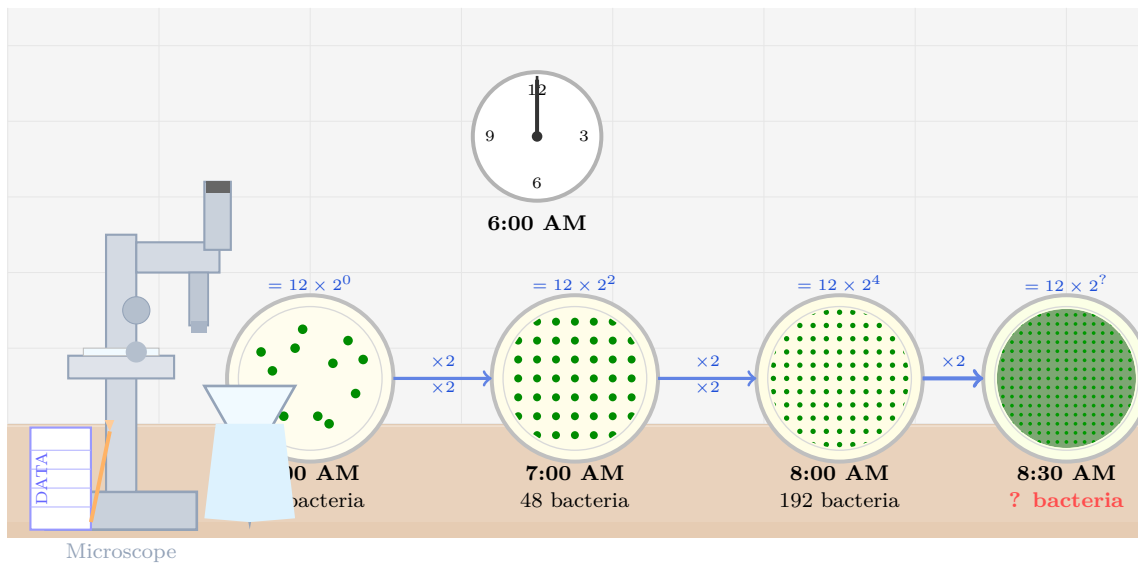
- What is the total depth of the drill bit after both shifts? Express your answer as a mixed number.
- What is the depth of the pressure sensor? Express your answer as a mixed number.

(a)

(b)

Problem 3

Powers & Exponents

**Problem 3: Bacterial Explosion**

A biology student places **12 bacteria** in a culture dish at **6:00 AM**. The bacteria double every **30 minutes**.

- How many doubling periods occur between 6:00 AM and 8:30 AM?
- Write an expression using a **power** (exponent) for the number of bacteria at 8:30 AM.
- Evaluate your expression. How many bacteria are in the dish at 8:30 AM?

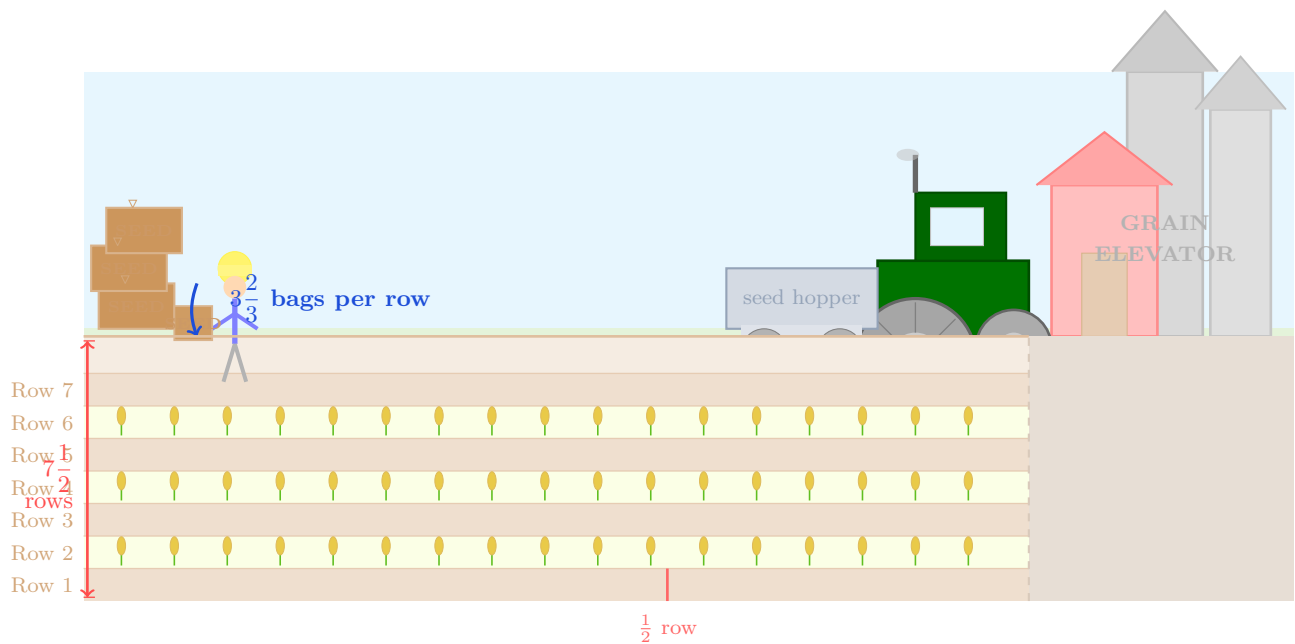
(a)

(b)

(c)

Problem 4

Rational Numbers

**Problem 4: Seeding the Field**

A farmer is seeding a wheat field. Each row of the field requires exactly $3\frac{2}{3}$ bags of seed. The field has $7\frac{1}{2}$ rows to plant.

- Write a multiplication expression for the total number of bags of seed needed. Convert both mixed numbers to improper fractions and solve. Express your answer as a mixed number.
- Bags of seed cost \$28.50 each. If the farmer can only purchase whole bags, what is the total cost to buy enough seed to plant the entire field?

(a)

(b)
